ARCHITECTURE SELECTION AND JUSTIFICATION

Project: Remote Patient Vitals Alert & Monitoring App

Problem Statement: #16

Student SRN: PES1UG24CS444

Selected Architecture: Microservices Architecture

Architecture Selection

Microservices Architecture was chosen for the Remote Patient Vitals Alert & Monitoring App. The system is divided into independently manageable services for telemetry ingestion, vital monitoring and detection, clinical threshold management, alert generation and escalation, caregiver interaction, and emergency dispatch integration.

Reasons for Selection

1. Independent scaling for continuous telemetry processing

The system must support at least 500 concurrent continuous telemetry streams while continuously evaluating patient vitals. Separating telemetry ingestion and vital monitoring into dedicated services allows these functions to be scaled independently as the number of telemetry streams increases.

2. Fault isolation for critical alert handling

The application contains time-critical alert and escalation functions. Separating alert generation and escalation from other services reduces the impact of failures in non-critical components and allows critical alert processing to remain independently managed.

Security Advantage

Microservices provide separate service boundaries where authentication and authorization controls can be enforced. Access to sensitive patient information can be restricted to authorized services and users, while communication between services can use encrypted connections. This supports the requirement for protected patient data, encryption, and role-based access control.

Performance Benefit

Telemetry ingestion and anomaly detection can operate independently of the caregiver dashboard. Incoming vital readings can therefore be evaluated and critical anomalies processed without depending on dashboard interaction. This supports the requirement to identify critical anomalies within 2 seconds. Actual latency and throughput would need to be verified through implementation and testing.